Jae-Won Jang

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EDUCATION

Doctor of Philosophy Student in Computer Engineering (2018-), Electrical and **Computer Engineering,** Virginia Tech
MS, Computer Engineering, University of South Florida (2015) GPA: 3.76 / 4.00
Thesis Title: Security of Non-Volatile Memories – Attack Models, Analyses, and Counter-Measures
BS, Computer Engineering, University of South Florida (2013) GPA: 3.69 / 4.00 Engineering GPA: 3.83 / 4.00

EXPERIENCE

Summer 2013, **Intern**, Intel Corporation, Hillsboro, OR

I was a 2013 Intel Summer Intern. My main task was researching and integrating the AIMSuite API with a project called Shelf Edge Technology (SET). I was responsible for debugging a crashing problem that existed with SET Android application. Moreover, I coded a Java program that utilized the AIMSuite API to output HTC Evo camera data into a table in real time. This utilized the Data Collection Framework (DCF) API and Cloud Servers for an immediate dashboard of AIMSuite Analytics.

Fall 2013 - Summer 2016, University of South Florida **Research Assistant**, - Tampa, FL
At USF, my research focused on employing CMOS and emerging memory technology.

- Developed the 7-T SRAM Physically Unclonable Function (PUF) using Magnetic Tunnel Junction (MTJ) and showed improved stability and robustness using HSpice, VerilogA, and Matlab.
- Designed layouts (utilized up to M8 layer) of non-volatile sequential elements and schematics using 65nm library Modeled the STT-RAM using Objective Oriented Micromagnetic Framework (OOMMF) software to analyze the vulnerabilities and simulate the impact of different form of attack.

Spring 2014 to Spring 2015, University of South Florida **INTO Tutor**, Tampa, FL

I tutored students on graduate-level Computer Engineering core courses (Operating Systems, Algorithm, and Computer Architecture) for the USF INTO pathway system.

Fall 2016 to Summer 2017, Penn State University **Research Assistant**, State College, PA

Researched on hardware security projects involving reverse engineering and gate camouflaging techniques.

Fall 2017, Penn State University **Teaching Assistantship**, State College, PA

Working as an instructor for the course CMPSC 122 – Intermediate Programming (Python data structure course)

February 2018 to August 2018, Raytheon **Software Engineer**, State College, PA

August 2018 to present, Virginia Tech **Research Assistant,** Blacksburg, VA

Working on the Popcorn Linux project (http://popcornlinux.org/) which aims to improve the programmability of emerging heterogeneous hardware.

SKILLS

- Programming Language: **C**, **C++**, **Java**, Javascript, **Matlab**, **Objective-C**, Perl, **Python**
- Database-related: Memcached. Redis, SQL
- IDE: Visual Studio, NetBeans, Eclipse, Github, Xilinx ISE, and SimpleScalar
- Proficient at using GNU / Linux distribution. Knowledgeable at Computer Architecture

RESEARCH INTEREST

Systems Engineering, Computer Architecture, Computer Security, Compilers, Linking

PUBLICATIONS

JOURNALS

1. **Jae-Won Jang**, Asmit De, Deepak Vontela, Ithihasa Nirmala, Swaroop Ghosh, and Anirudh Iyengar, "Threshold-defined Logic and Interconnect for Protection against Reverse Engineering", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**) (Impact Factor: 2.09)
2. Anirudh Iyengar, Swaroop Ghosh, and **Jae-Won Jang**, "MTJ-based State Retentive Flip-Flop with Enhanced-Scan Capability to Sustain Sudden Power Failure", Transactions on Circuits and Systems I (**TCAS-I**). (Impact Factor: 2.3)
3. Swaroop Ghosh, Anirudh Iyengar, Seyedhamidreza Motaman, Rekha Govindaraj, **Jae-Won Jang**, Jinil Chung, Jongsun Park, Xin Li, Rajiv Joshi, and Dinesh Somasekhar, "Overview of Circuits, Systems, and Applications of Spintronics", IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS). (Impact Factor: 1.6)
4. Anirudh Iyengar, Kenneth Ramclam, Swaroop Ghosh, **Jae-Won Jang** and Cheng-Wei Lin, "Spintronic PUFs for Security, Trust and Authentication", ACM Journal of Emerging Topics Computing Systems (**JETC Special Issue**). (Impact Factor 0.83)

CONFERENCES:

1. **Jae-Won Jang**, Jongsun Park, Swaroop Ghosh, and Swarup Bhunia, "Self-Correcting STTRAM under Magnetic Field Attacks", Design Automation Conference (**DAC**), 2015.
2. **Jae-Won Jang** and Swaroop Ghosh, "Performance Impact of Magnetic and Thermal Attack on STTRAM and Low-Overhead Mitigation Techniques", The International Symposium on Low Power Electronics and Design (**ISLPED**), 2016. Acceptance Rate = ~25%
3. **Jae-Won Jang** and Swaroop Ghosh, "Design and Analysis of Novel SRAM PUFs with Embedded Latch for Robustness", International Symposium on Quality Electronic Design Conference (**ISQED**), 2015.
4. **Jae-Won Jang,** Asmit De and Swaroop Ghosh, "Recent Trends in Intellectual Property (IP) Protection from Reverse Engineering", Microprocessor Test and Verification Conference (MTV), 2016 (Invited Paper).
5. **Jae-Won Jang** and Swaroop Ghosh, "Security of Magnetic Memories: Vulnerabilities, Attack Models, Analyses, and Prevention", **SRC TECHCON** 2015.
6. **Jae-Won Jang** and Swaroop Ghosh, "Improving Robustness of STTRAM Under Magnetic and Thermal Attack", **SRC TECHCON** 2016.
7. **Jae-Won Jang,** Deepak Reddy Vontela, Ithihasa Reddy Nirmala, and Swaroop Ghosh, "A Novel Threshold Voltage Defined Multiplexer for Interconnect Camouflaging", Government Microcircuit Applications and Critical Technology (**GOMATech-17**), 2017.
8. **Jae-Won Jang** and Swaroop Ghosh, "Security of Magnetic Memories: Vulnerabilities, Attack Models, Analyses and Preventions", Government Microcircuit Applications and Critical Technology (**GOMATech-16**), 2016.
9. Anirudh Iyengar, Deepak Vontela, Ithihasa Reddy, Swaroop Ghosh, Syedhamidreza Motaman, and **Jae-Won Jang "**Threshold Defined Camouflaged Gates in 65nm Technology for Reverse Engineering Protection", Proceedings of the International Symposium on Low Power Electronics and Design (**ISLPED**), 2018
10. Cheng-Wei Lin, **Jae-Won Jang,** and Swaroop Ghosh, "Schmitt-Trigger-Based Recycling Sensor and Robust and High-Quality PUFs for Counterfeit IC Detection", Government Microcircuit Applications and Critical Technology (**GOMATech-15**), 2015.
11. Alexander Holst, **Jae-Won Jang** and Swaroop Ghosh, "Investigation of Magnetic Field-Based Attacks on Magnetoresistive Random-Access Memory", International Symposium on Quality Electronic Design Conference (**ISQED**), 2017.
12. Swaroop Ghosh, Nasim Khan, Asmit De, **Jae-Won Jang**, "Security and Privacy Threats to On-Chip Non-Volatile Memories and Countermeasures", Proceedings of the 35th International Conference on Computer-Aided Design (**ICCAD '16**), 2016.

WORKS IN PROGRESS/DEMONSTRATIONS:

1. **Jae-Won Jang** and Swaroop Ghosh, "Data Integrity of STTRAM Vulnerabilities, Attack Models and Preventions", Design Automation Conference (**DAC**), 2016 (WIC).
2. Alexander Holst, Swaroop Ghosh, and **Jae-Won Jang,** "Investigation of Magnetic Field-Based Attacks on Magnetoresistive Random-Access Memory (Demo Proposal)", IEEE International Symposium on Hardware Oriented Security and Trust (**HOST**), 2016.
3. Anirudh Iyengar, Kenneth Ramclam, **Jae-Won Jang** & Cheng Wei Lin, Spintronic PUFs for Security, Trust and Authentication, Cyber Security Awareness Week Conference (CSAW), 2014. (3rd Place)

INVENTION DISCLOSURES

1. Non-Volatile Flip-Flop with Enhanced-Scan Capability to Sustain Sudden Power Failure, Swaroop Ghosh, Anirudh Iyengar, and **Jae-Won Jang,** United States patent US 9,728,241.2017 Aug 8.

RESEARCH EXPERIENCES

- Compilers
  - Research focusing on leveraging LLVM to just-in-time to different architecture at the run-time.
- Security of Non-Volatile (NV) Memories
  - Research focused on the exploration of Spintronics for the new type of PUF [J2]
  - Demonstrated the novel vulnerabilities of emerging memory technology (STT-RAM) [C2] using OOMMF.
  - Explored the improved robustness of hardware security by implementing MTJ onto SRAM PUF [C1, C4].
- Energy-Efficient Sequential Circuit Design
  - Explored reverse engineering [J1]
  - Employed Spintronics for energy-efficient circuit [J3]
- IC Design
  - Completed 1 tapeout. Experienced in PadFrame design using standard cmos065 library

CLASS PROJECTS

- Linux Kernel Course
  - Studied Linux Kernel throughout the semester and have an experience modifying Linux Kernel itself.
- Foundation of Software Security Project
  - Extensively studied **Buffer-Overflow Attacks** and implemented it on iOS. Demonstrated such attack on an iTouch 4th generation using Objective-C language.
- Emerging Topics in Network Security Project
  - Wrote a paper in regards to biometric authentication methods titled "Using User's Voice as the Catalyst for Biometric Authentication".
  - **Voice verification system** was researched upon and have implemented many different classifiers such as Gaussian Mixer Model (GMM), Hidden Markov Model (HMM), Fast Fourier Transform (FTT), and Dynamic Time Warp (DTW) to compare the effectiveness.
- CMOS Design Project
  - Designed, simulated, verified, and extracted IC layouts of basic gates, full adders, multiplexers, comparators, decoders, them using Virtuoso, Calibre, and DRC.
  - Designed thermometer semiconductor chip on the PadFrame using Virtuoso.
- Computer System Design Project
  - Programmed digital voice recorder using a ML505 FPGA board using Verilog language and MIPS Assembly Code. Recorder was able to record/play/pause/delete message.
- Senior Design Project
  - The goal of this project is to implement a new inventory management system for USF Sports Medicine – Athletic Department, which categorizes and organizes multiple items. Inventory management system is a system that can add or remove items, keeps track of all of the items, and manage items in the inventory. (usfinvent.com)

AWARDS, HONORS AND GRANTS

- HOST 2016 Student Grant
- Thomas E. and Ann C. Wade Scholarship

AFFILIATION AND LEADERSHIP

- Tau Beta Pi (National Engineering Honor Bachelor Society) Member